

OpenStax Chemistry 2e Errata

<https://openstax.org/details/books/chemistry-2e>

May 12, 2026 – Dean Peterson

Notes:

- I am a chemistry learner, not an expert, doing this avocationally not in a class setting.
- Since I lack instructor access, I could not review even numbered exercise solutions.
- I have attempted to include screenshots of unlabeled figures to assist in locating them.
- This document should have a PDF table of contents created from the headings.

Chapter 6

Chapter 6.1

Body text, section Waves, paragraph 3, sentence 1

Sentence 1 reads (emphasis mine):

All waves, including forms of electromagnetic radiation, are **characterized by, a wavelength** (denoted by λ , the lowercase Greek letter lambda), a frequency (denoted by ν , the lowercase Greek letter nu), and an amplitude.

I believe the third comma, the one between the phrases “characterized by” and “a wavelength”, is unnecessary and may not be grammatically correct.

Body text, section Waves, paragraph 3, sentence 3

Sentence 3 reads (emphasis mine):

Electromagnetic waves have wavelengths that fall within an enormous **range-wavelengths of** kilometers (10^3 m) to picometers (10^{-12} m) have been observed.

The punctuation between the word “range” and the phrase “wavelengths of” appears to be a hyphen.

(measured in meters in the SI
range-wavelengths of kilomet
of wave cycles that pass a sp

Shouldn't it be a dash of some sort? The hyphen makes it look like a hyphenated word. Compare with Ch 6.3, section Behavior in the Microscopic World, 1st sentence, where the punctuation between "macroscopic world" and "objects" appears to be an em dash, not a hyphen.

copic World
 macroscopic world—objects
 al physics. A billiard ball movir

Figure 6.2 alt text

Sentence 7 reads:

A bracket labeled, " λ_1 ," extends upward from the second and third peaks.

However, the bracket extends horizontally, not upward. I suggest changing sentence 7 to read:

A bracket labeled, " λ_1 ," extends from the second peak to the third peak.

Sentence 8 reads (emphasis mine):

Beneath this wave is the label, " ν_1 equals **4 cycles per second** equals 3 hertz."

The phrase "4 cycles per second" should read "**3 cycles per second**".

Sentence 10 starts (emphasis mine):

Beneath this wave is the label, " ν_2 equals **8 cycles per second** equals 6 hertz"

The phrase "8 cycles per second" should read "**6 cycles per second**".

There is no period at the end of sentence 10. In other words, the phrase:

... equals 6 hertz" The third wave ...

That should instead read:

... equals 6 hertz." The third wave ...

Sentence 14 (assuming a period was added for the preceding issue) reads (emphasis mine):

In the second column, **the** two waves are similarly shown, but lack the lambda labels.

In that sentence, the first word "the" appears superfluous.

Finally, the Figure 6.2 alt text does not mention the amplitude brackets labeled "a" on the waves in the second column. The preceding body text makes the amplitude definition of half the height from trough to peak sound important, so describing those brackets may be worthwhile.

Figure 6.3 alt text

I am seriously impressed with the clarity and detail of the alt text for this very information dense figure. I did notice a couple of issues though.

In sentence 11, the microwave clause reads (emphasis mine)

... microwave **infrared** appears in the 10^{10} superscript negative 2 to 10^{10} superscript negative 1 range ...

The word “infrared” should be removed from this location. It had its own clause earlier.

The last 3 sentences read:

Above broadcast and wireless radio are two images. The left most image is a black and white medical ultrasound image. A wireless AM radio is positioned at the far right in the image, also above broadcast and wireless radio.

There is no ultrasound image in the diagram, so those sentences should read something like:

Above broadcast and wireless radio, at the far right, is an image of a wireless AM radio.

Figure 6.5 alt text

The alt text currently reads:

This figure shows 3 wave diagrams. The first wave diagram is in black and shows two crests, indicates a consistent distance from peak to trough, and has one trough in its span across the page. The label, “Signal,” appears to the right. Just below this, a wave diagram is shown in red. The wave includes sixteen crests, but the distance from the peaks to troughs of consecutive waves varies moving across the page. The peak to trough distance is greatest in the region below the peaks of the black wave diagram, and the distance from peak to trough is similarly least below the trough of the black wave diagram. This red wave diagram is labeled, “A M.” The third wave diagram is shown in blue. The distance from peak to trough of consecutive waves is constant across the page, but the peaks and troughs are more closely packed in the region below the peaks of the black wave diagram at the top of the figure. The peaks and troughs are relatively widely spaced below the trough region of the black wave diagram. This blue wave diagram is labeled “F M.”

Without reference to the diagram, I think this text would have confused me. For example, take the phrase about the red AM wave which says “but the distance from the peaks to troughs of consecutive waves varies moving across the page”. It sounds unclear to me whether it is talking about vertical distance (proportional to amplitude) or horizontal distance (wavelength).

Here is an attempt to clarify it:

This figure shows 3 wave diagrams. All three waves run horizontally, varying up and down as they go from left to right. The first wave diagram is in black. It has a long wavelength, fitting only two crests and one trough in its span. It has consistent height from peak to trough. This black wave diagram is labelled, “Signal.” Just below this, a wave diagram is shown in red. The wave includes sixteen crests. The horizontal distance from one peak to the next is consistent across the entire wave. However, the vertical

height from the peaks to the troughs varies moving across the page. The peak to trough height is the greatest in the regions below the peaks of the black wave diagram. Conversely, the peak to trough height is the smallest below the trough of the black wave diagram. This red wave diagram is labeled, “A M.” The third wave diagram is shown in blue. The vertical height from the peaks to troughs is consistent across the entire wave. However, the horizontal distance from one peak to the next varies across the page. The peaks are more closely packed in the regions below the peaks of the black wave diagram at the top of the figure. Conversely, the peaks are relatively widely spaced below the trough region of the black wave diagram. This blue wave diagram is labeled “F M.”

Figure 6.6 alt text

The alt text refers to four color bands, when the image only shows one (red). Perhaps the image was updated but the alt text was not. The alt text needs to be rewritten.

Figure 6.6 caption

The caption currently reads:

Interference fringe patterns are shown for light passing through two closely spaced, narrow slits. The spacing of the fringes depends on the wavelength, with the fringes being more closely spaced for the shorter-wavelength blue light. (credit: PASCO)

I think the second sentence can be removed. It appears to be referring to different interference patterns for multiple colors, but the image now only shows interference patterns for one color, red.

Figure 6.7 alt text

The alt text currently reads (emphasis mine):

This figure shows four one-dimensional standing waves. The waves are shown in a tan color and are composed of curves to represent standing waves that can be generated using string. The first image at the top of the figure shows a single long wave with no nodes, or points where the string appears to cross between the endpoints at the left and right sides of the figure. The second diagram just below shows a single node at the center of the wave, which divides the **wave** into two identical halves to the left and right. The third diagram shows two nodes, dividing the **image** into three identical parts to the left, center, and right. Similarly, the last image at the bottom of the figure shows three nodes, dividing the **image** into four identical parts.

To me, it is not clear from the text that the endpoints of the waves are pinned and held fixed. Also, the language in the last two sentences changes from “divides the wave” to “dividing the image”. Continuing the “dividing the wave” language seems more appropriate.

Suggested alt text with changes in bold:

This figure shows four one-dimensional standing waves. The waves are shown in a tan color and are composed of curves to represent standing waves that can be generated using string. **The tan string is held fixed at both ends.** The first image at the top of the figure shows a single long wave with no nodes, or points where the string appears to cross between the endpoints at the left and right sides of the figure. The second diagram just below shows a single node at the center of the wave, which divides the **wave** into two identical halves to the left and right. The third diagram shows two nodes, dividing the **wave** into three identical parts to the left, center, and right. Similarly, the last image at the bottom of the figure shows three nodes, dividing the **wave** into four identical parts.

Figure 6.8 alt text

Sentence 3 is describing the elements common to all four pictures, but gives a specific frequency readout value that is not applicable to all four pictures:

To the right of the platform is a white box with a blue front which is labeled, "5.289 k H z Function Generator."

A slightly more accurate replacement would be:

To the right of the platform is a white box with a blue front which is labeled, "Function Generator," and has a readout showing a number of k H z.

Figure 6.9 alt text and caption

The caption calls the color of the blackbody spectrum line "blue", while the alt text repeatedly calls the color of the blackbody spectrum line "grey". It might be good to make them consistent. (It looks blue to me, but I am partially colorblind.)

Figure 6.10 alt text

It appears the diagram changed since this alt text was written. Issues include:

- Horizontal units are nm in the image but micrometers in the alt text. This issue applies to both the X axis label and everywhere a location on the X axis is described.
- The vertical axis label is not exactly correct. No capital letter I and the word arbitrary is not abbreviated.
- One X axis location is given as 7.2 micrometers, which would have been incorrect even if the X axis was in micrometers. It likely should have been 0.72 micrometers and would therefore now be 720 nm.
- The temperature values given for the colored curves do not match the temperature values on the image.

The alt text needs to be rewritten.

Figure 6.13 alt text and Exercise 6.30 image alt text

The images and alt text for these two are identical, so the following comments apply to both.

Sentence 2 reads (emphasis mine):

Across the top and bottom of the image is a scale that begins at 4000 angstroms at the left and extends to **740** angstroms at the far right.

The number 740 should be 7400.

The alt text describes the emission spectra as containing vertical bands, but never specifies that most of the spectrum is dark and that the vertical bands are light on dark, which the visual image makes obvious. This could be addressed by inserting a new sentence 4, just before describing row 2, which reads:

The remaining rows show emission spectra, which are mostly dark but contain narrow bands of color.

Chapter 6.2

Figure 6.14 alt text

The final sentence reads:

Just a short distance above this segment, a brown horizontal line segment is drawn. A black line segment extends to the right to the label, "0.00 J, infinity."

On the visual diagram, the number is 0.0 J, not 0.00 J.

Chapter 6.3

Figure 6.16 alt text

The current alt text reads:

A photograph is shown of ripples in water. The ripples display an interference pattern with each other.

That does not seem to capture what the ripples or the interference patterns look like, and adds no information to the caption. Perhaps this would be a bit better?

A photograph is shown of ripples in water. Circular ripples spread out in concentric rings from multiple points on the surface of the water. They form interference patterns where the ripples spreading out from different centers meet each other, reinforcing or cancelling each other in complicated patterns.

Figure 6.20 alt text

The alttext currently reads:

This figure provides images and graphs to illustrate the probability of finding an electron in 1 s, 2 s, and 3 s orbitals as a function of the distance from the nucleus. The 1 s orbital is shown as a sphere with a chunk missing. The graph below it has x axis labeled “distance from nucleus” and y axis labeled “probability density”. The related curve quickly reaches a maximum height and rapidly declines. The label, “1 s” appears below the graph. The 2 s orbital is shown as a red sphere with a blue middle. A chunk is missing from the sphere. The graph below it has x axis labeled “distance from nucleus” and y axis labeled “probability density”. The related curve quickly reaches a relative maximum height, a significantly higher absolute maximum height, and then rapidly declines. The label “2s” appears below it. The 3 s orbital is a blue sphere with a red sphere and another blue sphere at its core. The graph below it has x axis labeled “distance from nucleus” and y axis labeled “probability density”. The related curve quickly reaches a relative maximum height, a second relative maximum height, a significantly higher absolute maximum, and then declines more gradually than illustrated in the previous 2 graphs. The label, “3 s,” appears below the graph.

There are three issues with this alt text as currently written:

1. When sentence 2 says “The 1 s orbital is shown as a sphere with a chunk missing”, it is not clear that the chunk missing is merely a visibility aid to see inside the sets of spheres. The last sentence of the main text above the figure states “The number of radial nodes in an orbital is $n - l - 1$.” For the 1s orbital these numbers are $1 - 0 - 1 = 0$. In other words, there is no 0 probability region in the 1s orbital, but the alt text almost makes it sound like there is a 0 probability region in the 1s orbital by describing a “chunk missing.”
2. The 2s and 3s orbital sphere descriptions in sentences 6 and 11 do not mention that there is some space between the outside of one sphere and the inside of the next sphere. In other words, it doesn’t describe the radial nodes where the probability of finding an electron is at or near 0.
3. The 2s and 3s curve descriptions in sentences 9 and 13 say that the curve drops off after each maximum in the series, but it does not state that the curve drops off all the way to zero between each maximum, which again is a way of communicating the radial nodes.

Here is a possible suggestion for improved alt text:

This figure provides images and graphs to illustrate the probability of finding an electron in 1 s, 2 s, and 3 s orbitals as a function of the distance from the nucleus. On the top are three dimensional drawings of the orbital spheres. Each drawing has a wedge cut out so you can see internal details. Below each drawing is a graph with the x axis labeled “distance from nucleus” and the y axis labeled “probability density”. The 1 s orbital is drawn as a solid sphere with no holes in it. The graph below it has a curve with one hump where it rises from zero, quickly reaches a maximum height, rapidly declines, and then slowly tails off. The label, “1 s” appears below the graph. The 2 s orbital is drawn as a wide red spherical shell with a solid blue sphere in the middle. There is a small spherical gap between the outer red sphere and the inner blue sphere. The graph below

it has a two hump curve where it quickly reaches an initial maximum, goes back down to zero, goes up to a much higher maximum, rapidly declines, and then slowly tails off. The label “2s” appears below the graph. The 3 s orbital is drawn as a wide blue spherical shell with a narrower red spherical shell inside it and a solid blue sphere at its core. Between each sphere is a spherical gap. The gap between the innermost sphere and the second sphere is smaller than the gap between the second sphere and the outermost sphere. The graph below it has a three hump curve where it quickly reaches an initial maximum, goes back down to zero, goes up to a higher second maximum, goes back down to zero, goes back up to its highest maximum, and then declines and tails off more gradually than the previous 2 graphs. The label, “3 s,” appears below the graph.

Figure 6.21 alt text

The shapes of the d and f orbitals are not described. I can understand why. However, is there some accessible online resource with full descriptions that could be mentioned?

Figure 6.23 alt text

The alt text currently reads:

This diagram has an upward pointing arrow at the left which is labeled, “B subscript 0.” To the right, two spheres are shown. The first has a gray square at the top labeled, “N,” and a second gray square at the bottom labeled, “S.” A curved arrow is pointing right across the surface of the sphere and a gray arrow points upward through the center of the sphere. This sphere is labeled, “Spin plus one-half, spin-up.” The sphere just to the right has a gray square above it labeled, “S,” and a gray square below it labeled, “N.” This sphere has a curved arrow on its surface that is directed to the left and a gray arrow through the center of the sphere that points downward. This sphere is labeled, “Spin negative one-half spin-down.”

Issues with this alt text.

- This alt text does not mention that the spheres have e^- written in them.
- The spin arrow descriptions are reversed. The left spin arrow description says that “curved arrow is pointing right across the surface of the sphere”, but it is not, the curved arrow is pointing to the right as it goes behind the sphere, which is the opposite spin direction. The right spin arrow description has a similar issue.
- The N and S gray squares for the left e^- sphere are described in terms like “at the top”, which leaves it ambiguous whether they are at the top of but within the e^- spheres, or at the top of the diagram. The “above it” phasing used for the right e^- sphere seems clearer.

Here is suggested updated alt text (changes in bold):

This diagram has an upward pointing arrow at the left which is labeled, “B subscript 0.” To the right, two spheres **labeled “e superscript minus”** are shown. The first has a gray square **above it** labeled, “N,” and a second gray square **below it** labeled, “S.” A

curved arrow is pointing **left** across the surface of the sphere **showing it would be spinning clockwise if viewed from above**. A gray arrow points upward through the center of the sphere. This sphere is labeled, "Spin plus one-half, spin-up." The sphere just to the right has a gray square above it labeled, "S," and a gray square below it labeled, "N." This sphere has a curved arrow **pointing right across the surface of the sphere, showing that it would be spinning counterclockwise if viewed from above**. A gray arrow **points downward** through the center of the **sphere**. This sphere is labeled, "Spin negative one-half, spin-down."

Chapter 6.4

Figure 6.24 alt text

The alt text currently reads:

A table entitled, "Subshell electron capacity," is shown. Along the left side of the table, an upward pointing arrow labeled, "E," is drawn. The table includes three columns. The first column is narrow and is labeled, "2." The second is slightly wider and is labeled, "6." The third is slightly wider yet and is labeled, "10." The fourth is the widest and is labeled, "14." The first column begins at the very bottom with a horizontal line segment labeled "1 s." Evenly spaced line segments continue up to 7 s near the top of the column. In the second column, a horizontal dashed line segment labeled, "2 p," appears at a level between the 2 s and 3 s levels. Similarly 3 p appears at a level between 3 s and 4 s, 4 p appears just below 5 s, 5 p appears just below 6 s, and 6 p appears just below 7 s. In the third column, a dashed line labeled, "3 d," appears just below the level of 4 p. Similarly, 4 d appears just below 5 p and 5 d appears just below 6 p. Six d however appears above the levels of both 6 p and 7 s. The far right column entries begin with a dashed line labeled, "4 f," positioned at a level just below 5 d. Similarly, a second dashed line segment appears just below the level of 6 d, which is labeled, "5 f."

Issues:

- The alt text says there are three columns, but there are four.
- The number of line segments in the dashed horizontal lines are significant, but are not described in the alt text.
- It is hard to understand the overall pattern of increasing energy levels. Perhaps listing them from bottom to top at the end, irrespective of column, would help.

Suggested (changes in bold):

A table entitled, "Subshell electron capacity," is shown. Along the left side of the table, an upward pointing arrow labeled, "E," is drawn. The table includes **four** columns. The first column is narrow and is labeled, "2." The second is slightly wider and is labeled, "6." The third is slightly wider yet and is labeled, "10." The fourth is the widest and is labeled, "14." The first column begins at the very bottom with a **single** horizontal line segment labeled "1 s." Evenly spaced **single** line segments continue up to 7 s near the top of the column. **The second column contains horizontal dashed lines with 3 segments each. A**

dashed line labeled, “2 p,” appears at a level between the 2 s and 3 s levels. Similarly 3 p appears at a level between 3 s and 4 s, 4 p appears just below 5 s, 5 p appears just below 6 s, and 6 p appears just below 7 s. **The third column contains horizontal dashed lines with 5 segments each.** A dashed line labeled, “3 d,” appears just below the level of 4 p **but just above the level of 4 s.** Similarly, 4 d appears just below 5 p **but just above 5 s.** 5 d appears just below 6 p **but above 6s.** Six d however appears above the levels of both 6 p and 7 s. **The far right column contains horizontal dashed lines with 7 segments each. It begins** with a dashed line labeled, “4 f,” positioned at a level just below 5 d **but just above 6 s.** Similarly, a second dashed line segment labeled, “5 f,” appears just below the level of 6 d, **but just above the level of 7 s.** **The order of all energy levels from bottom to top are 1 s, 2 s, 2 p, 3 s, 3 p, 4 s, 3 d, 4 p, 5 s, 4 d, 5 p, 6 s, 4 f, 5 d, 6 p, 7 s, 5 f, and 6 d.**

Figure 6.25 alt text

The alt text currently reads:

A light blue hemisphere is labeled H. At a location about midway between the center and outer edge of the hemisphere, a small yellow-orange sphere is shown that is labeled with a negative sign. To the right of this diagram is the electron configuration 1 s superscript 1. The superscript is shown in a small yellow-orange circle. This superscript is labeled, “Number of electrons in subshell,” and the s is labeled, “Subshell.”

This describes the yellow-orange shading behind the superscripted 1 on the right side, but it does not describe the light blue shading behind the letter s on the right side.

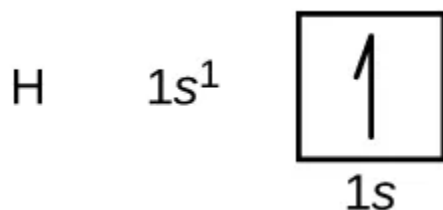
Here is an edit (changes in bold).

A light blue hemisphere is labeled H. At a location about midway between the center and outer edge of the hemisphere, a small yellow-orange sphere is shown that is labeled with a negative sign. To the right of this diagram is the electron configuration 1 s superscript 1. The superscript is shown in a small yellow-orange circle **and is** labeled, “Number of electrons in subshell.” The letter s is **shown in a small light blue circle and is** labeled, “Subshell.”

Figure 6.27 alt text

In the alt text of other figures, subshell names are given with a a space between the shell number and subshell letter, like “1 s”. However, the Figure 6.27 alt text contains subshell names with no space, such as “1s”, etc. I do not know if this is an issue, but I thought I’d mention it in case you usually include spaces because they help assistive technology.

Unlabeled H orbital diagram after Figure 6.27, alt text



This diagram is located after the first paragraph after Figure 6.27.

The first sentence reads (emphasis mine):

In this figure, the element symbol H is followed by the electron configuration **is** 1 s superscript 1.

I think the second word “is” in sentence 1 is ungrammatical. Here is a suggested replacement. It is patterned after the alt text for He orbital diagram below (change in bold):

In this figure, the element symbol H is followed by the electron configuration, “**1 s superscript 1.**”

Figure 6.29 alt text, issue

The alt text reads:

A periodic table, entitled, “Electron Configuration Table” is shown. The table includes the outer electron configuration information, atomic numbers, and element symbols for all elements. A square for the element hydrogen is pulled out beneath the table to provide detail. The blue shaded square includes the atomic number in the upper left corner, which is 1; the element symbol, H, in the upper right corner; and the outer electron configuration in the lower, central portion of the square. For H, this is 1s superscript 1.

Note that this description does not include anything about which regions of the periodic table are shaded in which color. However, the main body text later makes reference to the colors on Figure 6.29 as a distinction that matters. Specifically, Chapter 6.4, section Electron Configurations and the Periodic Table, 3rd through 6th paragraphs include (emphasis mine):

- The **colored sections of Figure 6.29** show the three categories of elements classified by the orbitals being filled: main group, transition, and inner transition elements.
- Main group elements (sometimes called representative elements) are those in which the last electron added enters an s or a p orbital in the outermost shell, **shown in blue and red in Figure 6.29.**
- Transition elements or transition metals ... However, the term is frequently used to refer to the entire *d* block (**colored yellow in Figure 6.29**), and we will adopt this usage in this textbook.
- Inner transition elements ... They are **shown in green in Figure 6.29.**

Therefore, I suggest adding section (or group number) and color information to the alt text for Figure 6.29.

Chapter 6.5

Figure 6.30 (b)

If Figure 6.30(b) is there just to show the general trend, it may be fine. If it is intended to reward close inspection, however, I think the sizes of the radius balls are off. I wonder if that diagram is either:

- Not the covalent radii, but a different radii.
- Not to scale as the Figure 6.30 caption says it is.

Reasons I wonder that include:

- Table 6.2 gives covalent radius for F as 64 pm and for At as 148 pm, more than double that of F. However, the visual radius difference does not match that scale. Moving the two next to each other in a graphics program shows At less than double the radius of F.



- Close examination, especially moving things around in a graphics program, shows some size relationships that look odd, including Be < B, Al < Si, and Ga < Ge.



Checking data from three sources show the Covalent radii relationships being opposite for those pairs of elements. Sources checked are:

- Figure 6.31 (although that figure does not specify what type of atomic radii it is showing).
- The site used in the Link to Learning link above. Following the link, choosing Covalent Radius to the right and Scatter plot above leads to a line graph.
<https://periodictable.com/Properties/A/CovalentRadius.html>
- Wikipedia atomic radii article.
[https://en.wikipedia.org/wiki/Atomic_radii_of_the_elements_\(data_page\)](https://en.wikipedia.org/wiki/Atomic_radii_of_the_elements_(data_page))
(although they do match the van der Waals radii relationships on Wikipedia, so maybe the diagram was compiled from some other measure of radii?)

Figure 6.30 caption

The caption ends with the phrase:

... and decrease across a period.

Would this be a little more explicit? I especially think of people whose main language reads from right to left instead of left to right like English.

... and decrease across a period **from left to right**.

Figure 6.30 alt text

Sentence 6 reads (emphasis mine):

The end points of this arrow connect to line segments that extend to the atomic **radii** below.

Shouldn't the word "radii" be "**nuclei**"?

Sentence 19 reads:

No spheres are provided for the noble or inert gas—group 18 elements.

This is not true. Spheres for the noble gases are included, and sentence 19 can be removed.

Figure 6.31 alt text

Sentence 8 begins:

Similarly, minima exist at the locations of noble or inert gases: H e, N e, A r, K r, X e, and R n, ...

However, Rn is not on the graph. At atomic number 86, Radon would be off the right end of the graph.

The last 2 sentences read:

The atomic radii for the alkali metals in picometers are: L i 167, N a 190, K 243, R b 265, and C s 298. The atomic radii of the noble or inert gases included in the graph in picometers are: H e 31, N e 38, A r 71, K r 88, and X e 108.

These numeric atomic radii are:

- Not written on the graph.
- Wrong. For example, the radius for Li is given as 167 pm, when the graph shows it is almost exactly 150 pm. The radius for Ne is given as 38pm, when the graph shows it above the 50 pm line.

I suggest just removing the last two sentences.

Example 6.12 - section Check Your Learning

The possible answers given are Ne and He. Wouldn't H also be an acceptable answer?

Figure 6.33 alt text

The graph includes labels of the periods along the top, which are not described in the alt text. Because sentence 3 describes the vertical lines dividing the periods, I suggest inserting a new sentence 4:

Between the vertical lines, periods are labeled along the top of the diagram, starting with Period 2.

The last two sentences read (emphasis mine):

Elements labeled on the graph and their associated ionization energies are as follows: H 1310, H e 2370, L i 520, B e 900, B 800, C 1090, N 1400, O 1310, F 1680, N e 2080, N a 490, **M g 730**, P 1060, A r 1520, K 420, Z n 910, A s 960, B r 1140, K r 1350, R b 400, C d 870, X e 1170, T l 590, and R n 1030.

Mg is not labelled on the graph. I don't think it matters, but since I noticed it I am mentioning it.

Figure 6.34 alt text

Holy canoli, that is impressive. Anyway, sentence 4 reads:

In group 2 the values are: B e 900, M g 730, C a 590, S r 550, and B a 500.

It is missing the value 510 for Ra at the end of the sentence.

Sentence 10 reads:

In group 8, the values are: F e 760, R u 720, and O s 840.

In the image, the value for Ru is 710.

The last sentence reads:

In group 18, the values listed are: B e 2370, N e 2080, A r 1520, K r 1350, X e 1170, and R n 1030.

I believe "B e" at the beginning is supposed to be "H e".

Figure 6.35 alt text

Sentence 3 reads:

Electron affinity values for representative elements are indicated with values marked with asterisks identifying calculated values.

Maybe it's just me, but I interpreted this as saying that representative elements would be marked with asterisks. I wonder if it would be clearer as two separate sentences.

Electron affinity values for representative elements are indicated. Values marked with asterisks identify calculated values.

Sentence 9 reads (emphasis mine):

In group 16, the values are: O negative 141, **S negative 20**, S e negative 195, T e negative 190, and P o negative 170.

According to the image, S negative 20 should be **S negative 200**.

Sentence 11 reads (emphasis mine):

In group 18, the values are: H e positive 20 asterisk, **N e negative 30**, A r positive 35 asterisk, K r positive 40 asterisk, X e positive 40 asterisk, and R n positive 40 asterisk. According to the image, Ne negative 30 should be **Ne positive 30 asterisk**.

Chapter 6 - Key Terms

frequency (ν)

Since the web display uses sans serif fonts, it is harder to tell the difference between an english letter “v” and the Greek letter nu. Would it make sense to replace **frequency (ν)** with **frequency (Greek letter ν)** or **frequency (ν - Greek letter nu)**?

Chapter 6 - Key Equations

The second equation for E gives the value for Planck’s constant h.

Would it make sense to likewise give the value of the Bohr radius a_0 after the last equation, the one for radius. For example, it might help students work exercise 6.25 without searching through chapter 6.2.

Chapter 6 - Exercises

Note: Only odd numbered exercises reviewed because I lack instructor access.

Exercise 6.7 solution

The book answer for the frequency part of the question is $\nu = 4.568 \times 10^{14}$ s. Shouldn’t the unit be inverted as per second, not second, and therefore be $\nu = 4.568 \times 10^{14} \text{ s}^{-1}$?

Exercise 6.9(a) solution

The book answer for (a) says the wavelength $\lambda = 8.69 \times 10^{-7}$ m, which is 869 nm, and I believe that is correct. However, the book also says that 869 nm electromagnetic radiation is “red”. I believe 869 nm is in the infrared part of the electromagnetic spectrum. This can be seen in Figures 6.9 and 6.10, both of which show the red end of the visible range ending around 700 nm. The Wikipedia articles on [Infrared](#) and [Visible spectrum](#) also agree that 869 nm is in the infrared, though they do disagree where the cutoff from visible to infrared is.

Exercise 6.11 solution

The energy part of the book answer is $E = 9.502 \times 10^{-15} \text{ J}$. However, using $c = 2.998 \times 10^8 \text{ m/s}$ and $h = 6.626 \times 10^{-34} \text{ J s}$, both from chapter 6.1, the following expression in python or google search bar gives a slightly different answer:

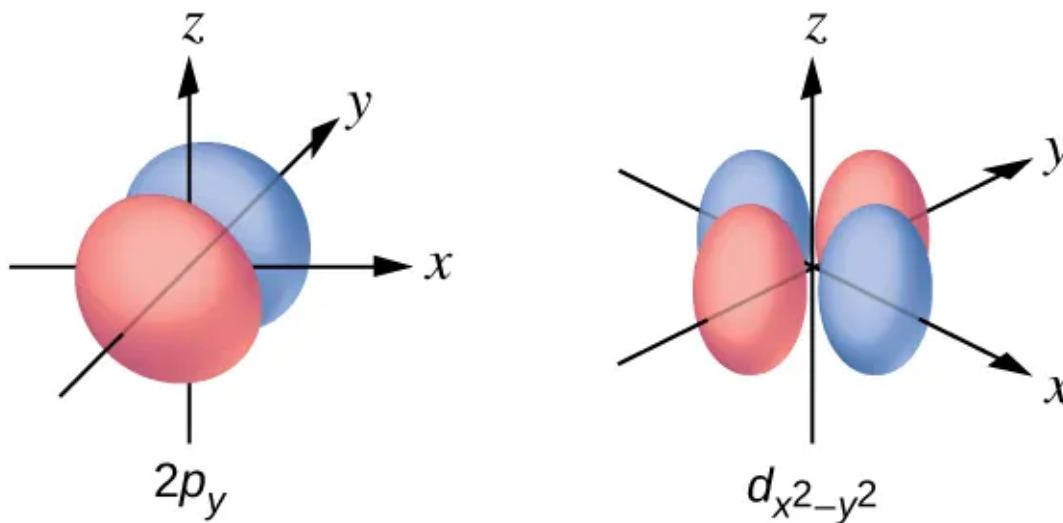
$$(6.626 * 10^{-34}) * (2.998 * 10^8) / (2.090 * 10^{-11}) = 9.504664 \times 10^{-15}$$

That rounds to $9.505 \times 10^{-15} \text{ J}$, not $9.502 \times 10^{-15} \text{ J}$. I even tried another calculator to be sure, so if my expression is correct, I think the book answer is slightly off.

Exercise 6.25 solution

This may not matter, but I'm wondering why the book answer, $33.9 \text{ \AA}$, is rounded to three significant digits when the Bohr radius input, $a_0 = 5.292 \times 10^{-11} \text{ m}$, is given to four significant digits in chapter 6.2. Wouldn't it be $33.87 \text{ \AA}$?

Exercise 6.39 solution image alt text



Sentence 5 reads:

The second diagram shows four ellipsoid lobes with ends centered around the origin. Are their ends centered around the origin? It looks more like their sides or, more to the point, their centers, are centered around the origin. "Ends centered around the origin" gives the impression of the long way of the ellipsoids pointing toward the origin, when in fact the long way is oriented up and down on the z axis.

Exercise 6.41(c) solution

The solution to exercise 6.13 includes the sentence “Somewhat different numbers are also possible.” Wouldn’t the same sentence be applicable to exercise 6.41 part (c)?

Exercise 6.51 solution

The book answer is “The charge on the ion.” That would certainly allow you to determine the ion, but aren’t there other possible answers as well? Specifically, wouldn’t any of the following allow you to determine the ion?

The charge on the ion or the name of the element or the atomic number of the element.

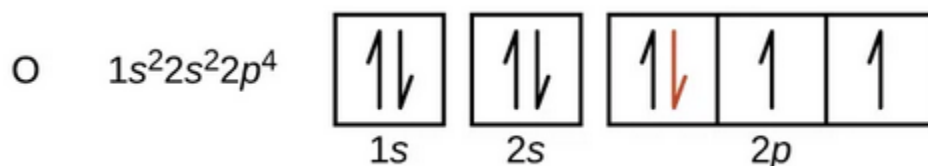
Exercise 6.53 solution sizing

The orbital diagrams in the 6.53 solution are small and rather hard to read. Here are three screenshots all taken at the same browser zoom of 100%.

From Chapter 6.4:

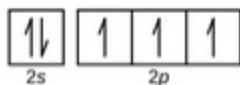


From Chapter 6.5. Not sure why it is smaller, but still easily readable:

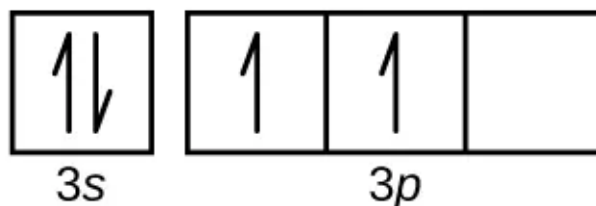


From the Exercise 6.53 Solution. (I believe this screenshot looks even bigger than the actual web page. The browser is still at 100% zoom. My 125% Windows display scaling may be feeding into the screenshot size. Regardless, you can see how much smaller it is compared to the others.)

53. (a)



Exercise 6.53 (b) solution image alt text

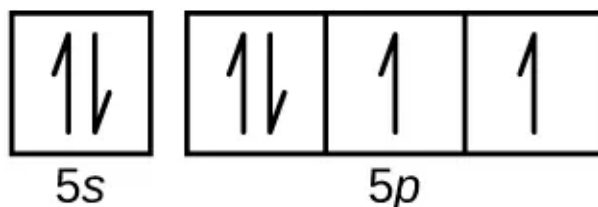


Sentence 2 reads:

The first square is labeled below as, “2 s.” The connected squares are labeled below as, “2 p.”

The “2 s” should be “3 s” and the “2 p” should be “3 p”.

Exercise 6.53 (d) solution image alt text



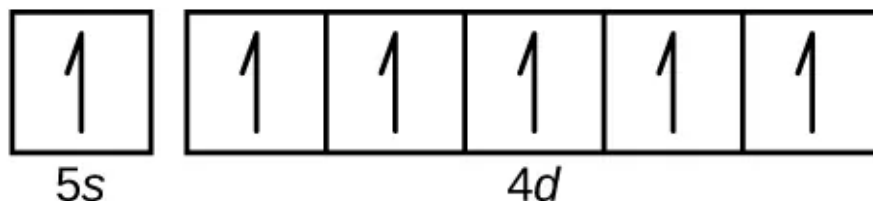
The alt text currently reads (emphases mine):

This figure includes a square followed by 3 squares all connected in a single row. The first square is labeled below as, “5 s ^{superscript 2}.” The connected squares are labeled below as, “5 p. ^{superscript 4}.” The first square and the left-most square in the row of **connect** squares each has a pair of half arrows: one pointing up and the other down.

Each of the remaining squares contains a single upward pointing arrow.

The yellow highlighted superscript phrases are not on the diagram and should be removed. The word “connect” should be corrected to “connected”.

Exercise 6.53 (e) solution image alt text



Sentence 3 currently reads:

The connected squares are labeled below as, “4 d ^{superscript 5}.” Each of the squares contains a single upward pointing arrow.

The yellow highlighted superscript phrase is not on the diagram and should be removed.

Exercise 6.63 and Exercise 6.65 questions

Questions 6.48, 6.49, and 6.54 start with text like:

Using complete subshell notation (not abbreviations, $1s^2 2s^2 2p^6$, and so forth), ...

Would it make sense to add similar verbiage to the questions of Exercise 6.63 and 6.65 as well?

The book solutions for Exercise 6.63 and 6.65 give the full electron configuration instead of starting with an abbreviation like [Xe]? Alternatively, would it make sense to add abbreviated versions like [Xe] $6s^2 4f^{14} 5d^{10}$ as alternate acceptable solutions?

Exercise 6.83 solution

The second part of the question reads:

What name is given to the energy for the reaction?

In addition to the book answer of IE_1 , would "First Ionization Energy" be an additional acceptable answer?